

Assignment - 02

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SET-5

1. Blowfish encryption steps

Blowfish is a symmetric block cipher using a 64-bit block and a variable key length.

It consists of:

- p-array
- Four S-boxes
- 16 Feistel rounds.

98/100



⇒ Step 1: Key Expansion

The user's key is used to initialize:

$p[0] \dots p[17]$

and four S-boxes.

The key is repeatedly XORed with the p-array.

⇒ Step 2: Initialize p-array and S-boxes

The p-array and S-boxes are initialized with predefined constants.

They are then modified using the encryption process.

⇒ Step 3: Divide plaintext

The 64-bit plaintext is divided into:

$L = 32$ bits.

$R = 32$ bits.

⇒ Step 4: 16 Feistel rounds

For each round i :

$L = L \text{ XOR } p[i]$

$$R = R \text{ XOR } F(L)$$

Then L and R are swapped.

→ Step 5: Final transformation

After round 16, the final swap is undone.

Then:

$$R = R \text{ XOR } p[16]$$

$$L = L \text{ XOR } p[17]$$

The two halves are combined to produce the ciphertext.

Blowfish F-Function

The 32-bit input is divided into four 8-bit parts:

a, b, c, d

Then:

$$F(x) = ((s_1[a] + s_2[b]) \text{ XOR } s_3[c]) + s_4[d]$$

Thus, S-boxes introduce non-linearity and provide strong confusion.

2. Diffie-Hellman - $p = 17, q = 3$

Given:

$$p = 17$$

$$q = 3$$

Alice private key $a = 6$

Bob's private key $b = 13$

Alice's public key

$$A = \alpha^a \text{ mod } q$$

$$B = \alpha^b \text{ mod } q$$

$$A = 3^6 \text{ mod } 17$$

$$A = 15$$

$$B = 3^{13} \text{ mod } 17$$

$$B = 12$$

shared Secret

Alice calculates:

$$K: B^a \bmod q$$

$$K: 12^6 \bmod 17$$

$$K: 2$$

Bob calculates:

$$K: A^b \bmod q$$

$$K: 15^{13} \bmod 17$$

$$K: 2$$

Therefore:

public key of Alice = 15

public key of Bob = 12

shared secret = 2

3. DES vs 3DES vs RSA vs ECC

Algorithm	Type	Security	Efficiency
DES	Symmetric	weak	Fast
3DES	Symmetric	Better than DES but obsolete	slow
RSA	Asymmetric	Strong with adequate key size	slower
ECC	Asymmetric	Strong	efficient

DES:

should not be used for modern security because its 56-bit key is too small.

3DES

Improves DES security by applying DES three times, but is slow and now considered legacy.

RSN

useful for digital signatures and public-key operations but requires relatively large keys and is computationally expensive.

ECC

provide a strong security with much smaller keys, making it suitable for mobile devices and IoT.

4. Stream vs Block ciphers

Block cipher

A block cipher processes plaintext in fixed-size blocks.

Example:

AES

File

1

128-bit Block1 → Encrypt

128-bit Block2 → Encrypt

128-bit Block3 → Encrypt

use case

- File encryption: AES can be used to encrypt documents, databases, and stored files.

stream cipher

A stream cipher produces a keystream and combines it with plaintext.

Plaintext XOR Keystream = ciphertext

use case

Real-time communication / video streaming:

Stream-oriented encryption is useful because data can be encrypted continuously without waiting for complete blocks.

Main difference

Block ciphers work on fixed-size blocks, whereas stream ciphers encrypt a continuous stream of data.

5. Quantum computing, RSA and post-Quantum cryptography

RSA security is based on the difficulty of factoring a very large integer into its prime factors.

Classical computers cannot efficiently factor sufficiently large RSA moduli.

However, a sufficiently powerful quantum computer could use Shor's algorithm to factor large integers efficiently.

Therefore:

Quantum computer → Shor's algorithm →

Efficient factoring → RSA becomes insecure

Large RSA number

↓

Quantum computer

↓

Shor's Algorithm

↓
Factorization

↓
RSA private key can be recovered.

post-Quantum cryptography

post-quantum cryptography (PQC) develops algorithms designed to remain secure against both classical and Quantum computers.

Major approaches include:

- Lattice-based cryptography
- Hash-based cryptography
- code-based cryptography
- Multivariate cryptography

The goal is to replace vulnerable public-key algorithms such as RSA and traditional Diffie Hellman with quantum resistant alternatives.

conclusion

Quantum computing creates a future threat to RSA and other classical public key systems. PQC is being developed to provide secure public-key encryption and digital signatures in the quantum

era.

